

Fig. S2. 18S rDNA-based identification and sequence comparison

A. BLAST-based identification of four isolates

M. pronini XY

Descriptions	Graphic Summary	Alignments	Taxonomy					
Sequences producing significant alignments								
Download		Select columns	Show 100					
☒ select all 100 sequences selected								
		GenBank	Graphics					
		Distance tree of results						
		MSA View						
Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒ Myxobolus pronini isolate ID8 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3507	100%	0.0	100.00%	1952	PP840605.1
☒ Myxobolus pronini isolate ID7 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3507	100%	0.0	100.00%	1959	PP840604.1
☒ Myxobolus pronini isolate ID1 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3507	100%	0.0	100.00%	1949	PP840598.1
☒ Myxobolus pronini isolate A0816 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3507	100%	0.0	100.00%	2034	MH329619.1
☒ Myxobolus pronini isolate S2 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3507	100%	0.0	100.00%	1952	QM678576.1
☒ Myxobolus pronini isolate S1 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3507	100%	0.0	100.00%	1943	QM678575.1
☒ Myxobolus pronini isolate TZ small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3507	100%	0.0	100.00%	1964	PZ492883.1
☒ Myxobolus pronini isolate ID6 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3502	3502	100%	0.0	99.95%	1994	PP840603.1
☒ Myxobolus pronini isolate ID2 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3502	3502	100%	0.0	99.95%	1992	PP840599.1
☒ Myxobolus pronini isolate XY small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3498	3498	100%	0.0	100.00%	1894	PZ492893.1
☒ Myxobolus pronini isolate ID3 small subunit ribosomal RNA gene and internal transcribed spacer 1, partial sequ...	Myxobolus pronini	3482	3482	100%	0.0	99.74%	2074	PP840600.1
☒ Myxobolus pronini isolate ID4 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3474	3474	100%	0.0	99.68%	2054	PP840601.1

M. pronini TZ

Descriptions	Graphic Summary	Alignments	Taxonomy						
Sequences producing significant alignments									
Download		Select columns	Show 100						
☒ select all 100 sequences selected									
Description		GenBank	Graphics	Distance tree of results	MSA Viewer				
	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession	
☒	Myxobolus pronini isolate ID8 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3577	100%	0.0	100.00%	1952	PP840605.1
☒	Myxobolus pronini isolate ID7 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3577	100%	0.0	100.00%	1959	PP840604.1
☒	Myxobolus pronini isolate ID1 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3577	100%	0.0	100.00%	1949	PP840598.1
☒	Myxobolus pronini isolate A0816 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3577	100%	0.0	100.00%	2034	MH329619.1
☒	Myxobolus pronini isolate S2 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3577	100%	0.0	100.00%	1952	OM678576.1
☒	Myxobolus pronini isolate S1 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3577	100%	0.0	100.00%	1943	OM678575.1
☒	Myxobolus pronini isolate TZ small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3577	100%	0.0	100.00%	1964	PZ492883.1
☒	Myxobolus pronini isolate ID6 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3502	3571	100%	0.0	99.95%	1994	PP840603.1
☒	Myxobolus pronini isolate ID2 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3502	3571	100%	0.0	99.95%	1992	PP840599.1
☒	Myxobolus pronini isolate XY small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3498	3568	100%	0.0	100.00%	1894	PZ492893.1
☒	Myxobolus pronini isolate ID3 small subunit ribosomal RNA gene and internal transcribed spacer 1, partial sequence	Myxobolus pronini	3482	3551	100%	0.0	99.74%	2074	PP840600.1
☒	Myxobolus pronini isolate ID4 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3474	3544	100%	0.0	99.68%	2054	PP840601.1

M. episquamalis QD

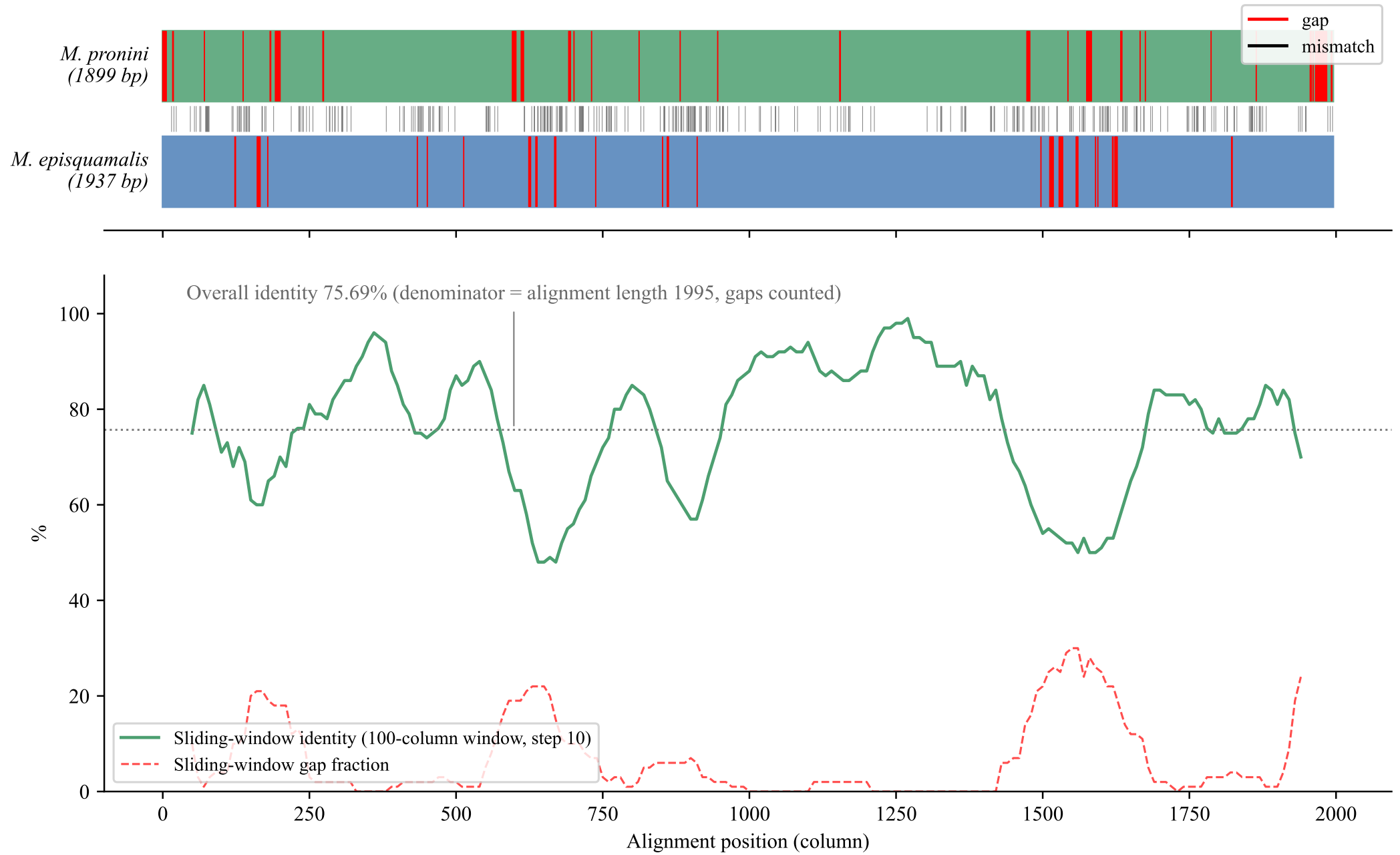
Descriptions	Graphic Summary	Alignments	Taxonomy					
Sequences producing significant alignments								
Download		Select columns	Show 100					
☒ select all 100 sequences selected								
GenBankGraphicsDistance tree of resultsMSA Viewer								
Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒ Myxobolus episquamalis isolate MG-qd-201908 small subunit ribosomal RNA gene, partial sequence	Myxobolus episq...	3578	3578	100%	0.0	100.00%	1937	QR621295.1
☒ Myxobolus episquamalis 18S ribosomal RNA gene, partial sequence	Myxobolus episq...	3474	3474	97%	0.0	100.00%	1881	JF810537.1
☒ Myxobolus episquamalis 18S ribosomal RNA gene, partial sequence	Myxobolus episq...	3452	3452	97%	0.0	99.84%	1885	MK329248.1
☒ Myxobolus episquamalis 18S ribosomal RNA gene, partial sequence	Myxobolus episq...	3195	3195	90%	0.0	99.77%	1742	AY129312.1
☒ Myxobolus patriciae isolate MC_01_I small subunit ribosomal RNA gene, partial sequence	Myxobolus patric...	2510	2510	91%	0.0	92.39%	1800	PP926143.1
☒ Myxobolus sp. m DHV-2022 isolate MC01 small subunit ribosomal RNA gene, partial sequence	Myxobolus mace...	2508	2508	95%	0.0	91.49%	1820	ON175973.1
☒ Myxobolus sp. c DHV-2022 isolate MC03 small subunit ribosomal RNA gene, partial sequence	Myxobolus sp. c...	2449	2449	96%	0.0	90.49%	1861	ON175975.1
☒ Myxobolus sp. per SR-2018 voucher CIMo small subunit ribosomal RNA gene, partial sequence	Myxobolus sp. p...	2433	2433	100%	0.0	89.49%	1976	MK203080.1
☒ Myxobolus sp. mus SR-2018 voucher CrMm small subunit ribosomal RNA gene, partial sequence	Myxobolus sp. m...	2416	2416	100%	0.0	89.29%	2009	MK203075.1
☒ Myxobolus sp. isolate C13MbD_C12MbD small subunit ribosomal RNA gene, partial sequence	Myxobolus sp.	2407	2407	100%	0.0	89.19%	1985	QQ319166.1
☒ Myxobolus sp. c DHV-2022 isolate MC02 small subunit ribosomal RNA gene, partial sequence	Myxobolus sp. c...	2364	2364	91%	0.0	91.04%	1767	ON175974.1

M. episquamalis XM

Descriptions	Graphic Summary	Alignments	Taxonomy						
Sequences producing significant alignments									
Download		Select columns	Show 100						
☒ select all 100 sequences selected									
		GenBank	Graphics	Distance tree of results	MSA Viewer				
	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	Myxobolus episquamalis isolate MG-gd-201908 small subunit ribosomal RNA gene, partial sequence	Myxobolus episq...	3578	3578	100%	0.0	100.00%	1937	QR621296.1
☒	Myxobolus episquamalis 18S ribosomal RNA gene, partial sequence	Myxobolus episq...	3474	3474	97%	0.0	100.00%	1881	JF810537.1
☒	Myxobolus episquamalis 18S ribosomal RNA gene, partial sequence	Myxobolus episq...	3452	3452	97%	0.0	99.84%	1885	MK329248.1
☒	Myxobolus episquamalis 18S ribosomal RNA gene, partial sequence	Myxobolus episq...	3195	3195	90%	0.0	99.77%	1742	AY129312.1
☒	Myxobolus patriciae isolate MC_01_I small subunit ribosomal RNA gene, partial sequence	Myxobolus patric...	2510	2510	91%	0.0	92.39%	1800	PP926143.1
☒	Myxobolus sp. m DHV-2022 isolate MC01 small subunit ribosomal RNA gene, partial sequence	Myxobolus mace...	2508	2508	95%	0.0	91.49%	1820	ON175973.1
☒	Myxobolus sp. c DHV-2022 isolate MC03 small subunit ribosomal RNA gene, partial sequence	Myxobolus sp. c...	2449	2449	96%	0.0	90.49%	1861	ON175975.1
☒	Myxobolus sp. per SR-2018 voucher CIMp small subunit ribosomal RNA gene, partial sequence	Myxobolus sp. p...	2433	2433	100%	0.0	89.49%	1976	MK203080.1
☒	Myxobolus sp. mus SR-2018 voucher CrMm small subunit ribosomal RNA gene, partial sequence	Myxobolus sp. m...	2416	2416	100%	0.0	89.29%	2009	MK203075.1
☒	Myxobolus sp. isolate C13MBD_C112MBD small subunit ribosomal RNA gene, partial sequence	Myxobolus sp.	2407	2407	100%	0.0	89.19%	1985	QQ319166.1
☒	Myxobolus sp. c DHV-2022 isolate MC02 small subunit ribosomal RNA gene, partial sequence	Myxobolus sp. c...	2364	2364	91%	0.0	91.04%	1767	ON175974.1

B. Pairwise 18S rDNA alignment and sequence identity

Needleman-Wunsch global alignment: match +2; mismatch -1; gap opening -5; gap extension -0.5.



Alignment: 1,995 columns Matches: 1,510 Mismatches: 331 Gap columns: 154

Sequence identity: 75.69% including gap columns (1,510/1,995); 82.02% excluding gap columns (1,510/1,841).